SABHARISH VARSHAAN S

Backend Engineer | FastAPI, PostgreSQL
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Education

SRM Institute of Science and Technology Jun 2023 – May 2027
B.Tech in Computer Science and Engineering **CGPA: 9.48/10**

Technical Skills

Backend & APIs: Python, FastAPI, PostgreSQL, Redis, REST APIs, JWT Authentication, RBAC, Pydantic, SQLAlchemy, SQL
Full-Stack & Automation: Supabase, APScheduler, Git, GitHub
Also worked with: Gunicorn, FAISS, Ollama/Mistral

Projects

Smart Attendance Platform FastAPI, PostgreSQL, Redis, React, Docker
Live Website | GitHub Repository

- Deployed a role-based attendance platform serving **2,500+ students** across four interfaces (student, teacher, incharge, admin), replacing manual Excel tracking.
- Built **66 REST APIs** with FastAPI and PostgreSQL, sustaining p99 latency under 1s at peak concurrent load, backed by 23 performance indexes (15 single-column, 8 composite) tuned to real dashboard query patterns.
- Resolved a race condition in concurrent attendance updates using atomic database-level operations, and built a custom Redis sliding-window rate limiter tuned for shared campus WiFi/NAT conditions.
- Secured the platform with JWT, bcrypt, and RBAC; migrated auth storage from localStorage to HttpOnly cookies after identifying and fixing an XSS vulnerability.

GuardGig FastAPI, PostgreSQL, Supabase, Scikit-Learn, APScheduler
GitHub Repository

- Built an automated insurance claims backend with a service-layer architecture separating routes from business logic, across policy, claims, and payout workflows for policyholders, agents, and admins.
- Implemented a scheduled job (APScheduler) running every 5 minutes to evaluate live trigger data against active policies, automatically generating and processing eligible claims.
- Secured endpoints with JWT authentication and RBAC, using a stateful validation design to support immediate access revocation.
- Integrated a Scikit-Learn model to flag high-risk claims for fraud review, prioritizing cases for manual investigation.

Preop-Triage FastAPI, Next.js, TypeScript, FAISS, Ollama/Mistral, SQLAlchemy
GitHub Repository

- Built a clinical decision-support system combining a deterministic rules engine with an AI-assisted advisory layer for pre-operative risk scoring.
- Designed role-based workflows (patient/doctor/admin) with JWT authentication, automated case routing for moderate/high-risk patients, and a validation layer to catch unsafe AI-generated content before it reaches a user.

Achievements & Co-Curricular Activities

- **Technical Lead, Intellects Club** Jun 2025 – Mar 2026
  Organized 3 workshops/competitions on modern technical trends as part of a 60-member technical club, each with 50–70 attendees.
- **Development Team Member, Placement Attendance Management System**
  Built a core module of an attendance system used to track student participation in placement training.